

Software Requirements Specification

Project: Topic-Based Daily News Digest Subscription Service

Version: 1.0

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Status: Model Answer / Approved

Revision History

Version	Date	Author	Change Summary	Approval
1.0	31-08-2025	Instructor	Initial SRS with diagrams added	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall Description
3. External Interfaces

4. System Features (Detailed)
 5. Non-Functional Requirements (Detailed)
 6. Quality Attributes & Acceptance Tests
 7. UML Use-Case Diagram
 8. Requirements Traceability Matrix (RTM)
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1. Introduction

1.1 Purpose

This SRS defines functional and non-functional requirements, interfaces, and verification criteria for a subscription service that delivers daily topic-based news digests to users via email .

1.2 Scope

The service enables users to select news topics, stores their preferences, fetches and compiles digest content from selected news sources, and generates templated email digests sent daily. Excludes direct integration with real-time news APIs or actual email delivery systems beyond stubs/mocks.^[1]

1.3 Audience

Developers, QA Engineers, System Integrators, Maintenance Technicians, and evaluators.

1.4 Definitions

- Digest: Compiled email containing daily news
- Topic: Subject area selected by the user
- Source: News provider included in the digest
- Stub: Simulated function/component, not production
- Mock: Test double used for simulating external behavior
- API: Application Programming Interface
- UI: User Interface
- TLS: Transport Layer Security

2. Overall Description

2.1 Product Perspective

A web-based subscription platform storing topic preferences, scheduling and executing digest-compilation jobs, integrating with mock news source APIs, and generating email content using a templating engine.

2.2 Major Product Functions

- User registration and authentication
- Topic selection and preference storage
- Management of available news sources
- Daily digest compilation job
- Email templating and mock delivery
- Administrative functions
- Job monitoring and error logging
- Offline queuing for digests (if service down)^[4]

2.3 User Roles and Characteristics

- Subscriber: Wants digest with chosen topics, expects easy configuration and reliable daily delivery
- Administrator: Manages sources, monitors digests, reviews logs
- Technician: Troubleshoots system, reviews job executions/logs

2.4 Operating Environment

Web application running on cloud/in-house server, database (SQL/NoSQL), secure network, and scheduled job processor.

2.5 Constraints

- GDPR-compliant user data storage

- TLS used for all authentication and API endpoints
 - Modular API framework for digest/job/source integration
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3. *External Interface Requirements*

3.1 User Interfaces

- Web UI: Topic selection, source management, account settings, accessible for visually impaired
- Email UI: HTML/text digest template for daily emails (displayed in browser, not sent)

3.2 Hardware Interfaces

- None directly (standard server/cloud hardware)

3.3 Software Interfaces

- Mocked News Source API (JSON over HTTPS)
- Job Scheduler (e.g., cron/emulator)
- Email Templating Stub

3.4 Communications

- TLS 1.2+ for all APIs
 - Retry strategy for failed digests
 - Secure logging and archiving^[1]
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4. System Features (Detailed)

Requirements listed below use IDs SUB-F-###, with acceptance criteria and test refs.

4.1 User Management

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Comments
SUB-F-001	System shall allow new user registration and authentication	Functional	High	Business / Security	AC-SUB-F-001: User can register, login, see only own preferences	Uses standard authentication
SUB-F-002	System shall store and retrieve user topic preferences securely	Functional	High	Security / Business	AC-SUB-F-002: User can change/add topics and data is always encrypted in transit	Data privacy required
SUB-F-003	System shall allow users to manage their preferred news sources	Functional	Medium	Business	AC-SUB-F-003: Users can add/remove/modify sources via UI	Source whitelist
SUB-F-004	System shall allow users to reset password via registered email	Functional	Medium	User / Security	AC-SUB-F-004: User can reset password securely using email verification	Mandatory for account recovery
SUB-F-005	System shall log all user login attempts (successful and failed)	Functional	Medium	Security / Admin	AC-SUB-F-005: All login attempts visible in logs	Helps monitor security breaches
SUB-F-006	System shall allow users to unsubscribe or deactivate their account	Functional	Medium	User	AC-SUB-F-006: User can deactivate/unsubscribe and receive confirmation	GDPR compliance requirement

4.2 Digest Compilation

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Comments
SUB-F-010	System shall schedule and execute daily digest compilation jobs	Functional	High	Business	AC-SUB-F-010: Digests are generated by job scheduler at configured time	Scheduler logs jobs

SUB-F-011	Digest must include at least 3 news items per user-selected topic	Functional	High	User	AC-SUB-F-011: Digest contains required news per topic	Checks API/stubs
SUB-F-012	System shall use templating engine to generate email content	Functional	Medium	Business	AC-SUB-F-012: Digest uses template for formatting and branding	Must be easily configurable
SUB-F-013	System shall retry failed digest jobs up to 3 times automatically	Functional	High	Admin	AC-SUB-F-013: Failed jobs are retried; if still failed, error is logged	Improves reliability
SUB-F-014	System shall archive generated digests for at least 30 days per user	Functional	Medium	User	AC-SUB-F-014: Users can access past digests from archive	Provides user convenience

4.3 Email Delivery (Mocked)

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Comments
SUB-F-020	Emails generated shall be sent to mock delivery system and displayed in browser for evaluation	Functional	Medium	QA/Evaluator	AC-SUB-F-020: Daily email is logged and displayable via UI	No actual sending
SUB-F-021	System shall provide digest preview before final delivery	Functional	Low	User	AC-SUB-F-021: Preview option available in UI before digest generation	Enhances usability

4.4 Admin & Monitoring

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Comments
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SUB-F-030	System shall provide admin dashboard for monitoring jobs, reviewing errors, and managing sources	Functional	Medium	Admin	AC-SUB-F-030: Logs/jobs/source controls available in dashboard	Logs must be timestamped
SUB-F-031	System shall generate alerts/notifications for failed digest jobs	Functional	High	Admin/Technician	AC-SUB-F-031: Alerts generated when a digest job fails	Improves system monitoring

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Category	Priority	Acceptance Criteria
SUB-NF-001	95% of digest jobs shall complete within 10 seconds	Performance	High	95th percentile <= 10s on test
SUB-NF-002	System shall be available 99.5% monthly	Reliability	High	Uptime reports show >= 99.5% per month
SUB-NF-003	All user data must be stored encrypted at rest	Security/Compliance	High	Security audit pass
SUB-NF-004	Logs with events/errors retained for 3 years	Audit/Data Retention	Medium	Archival and retrieval verified
SUB-NF-005	Service shall support screen-readers and high contrast mode	Accessibility	Medium	Accessibility audit pass

5.1 Security

5.1.1 Security Objectives

- Protect user data and privacy
- Ensure reliable, authenticated API communication

5.1.2 Security Requirements

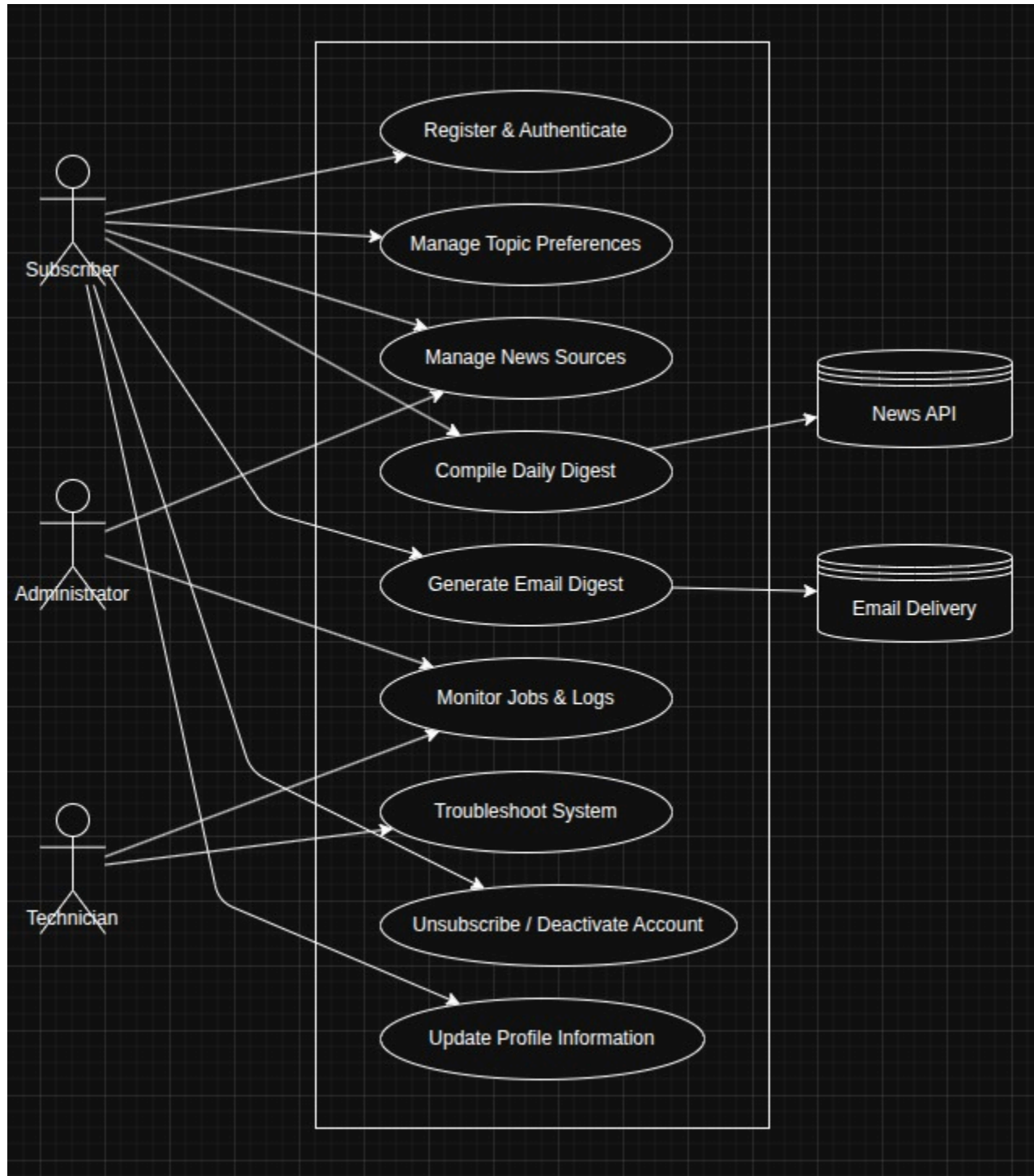
Req ID	Requirement	Type	Priority	Acceptance Criteria
SUB-SR-001	TLS 1.2+ mandatory for all communications	Security	High	All API/user communication passes TLS tests
SUB-SR-002	Only hashed passwords stored; no plain-text	Security	High	Penetration test pass
SUB-NF-003	All user data must be encrypted at rest	Security	High	Security audit verifies data encryption
SUB-SR-004	Regular vulnerability scans on APIs	Security	Medium	Scan log reviewed monthly
SUB-SR-005	Logs retained securely for 3 years with timestamping	Audit/Security	Medium	Archival & retrieval of logs verified

6. Quality Attributes & Acceptance Tests

- Exit criteria: All high-priority functional requirements implemented, verified, no critical NFR failures, RTM shows all test cases passed.
 - Acceptance test suites: User Management, Digest Compilation, Templating, Admin Dashboard, Performance, Security, Accessibility.^[1]
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7. System Models and Diagrams

UML use-case diagram



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref/Design Spec	Module	Test Case(s)	Statu s	Comment s
SUB-F-001	Register & Authenticate User	4.1 / DS-User-01	AuthModule	TC-User-01	N	
SUB-F-011	Digest includes 3+ items per topic	4.2 / DS-Digest-02	DigestModule	TC-Digest-02	N	
SUB-NF-001	Digest job $\leq$ 10s for 95% cases	5 / DS-Perf-01	JobScheduler	TC-Perf-01	N	
